

For A Successful VERTICAL FARMING Tech Business Put Farming First



Janani Gopalakrishnan Vikram is a technically-qualified freelance writer, editor and hands-on mom based in Chennai

If you wish to sell technology to verticals like healthcare and education, you have to learn to push technology to the background. What is more important is to understand the market, sync with its psyche and build a mutual trust with the customer base. Technology will automatically fall into place and earn you the due profits.

Well, the emerging space of vertical (indoor) farming is no different. Although a lot of exciting technology goes into controlling the environment and managing the crop, the key to succeeding in this business relies on how well you are able to forget that you are a technologist and focus on the farming requirements instead, so you can eventually provide the customer with the right amount of the right tech. Overdo it, and you will not get your next customer!

Getting down to business

Vertical farming seems to be a fast emerging business worldwide, especially in urban areas that yearn for safe, organic and fresh food. The modern notion

of vertical farming or controlled-environment farming involves growing crops and sometimes even fish and other food products in indoor spaces, using technology like LED lighting, heating, ventilation and air-conditioning (HVAC) systems, sensors and smart software to maintain total control over the environment.

"The Internet of Things (IoT), drones, mobile apps and social technologies can be brought together to create value concepts that empower farmers with timely information about crop cycles and health. This is pertinent in the context of indoor vertical farming since walking around to check how things are, is rarely an option. Data-driven knowledge, delivered in real time and based on location context is what farmers will need for making decisions in almost the same way that the Japanese made an art of just-in-time (JIT) manufacturing, since Japan lacked space to build big factories loaded with inventory," adds Sunil Malhotra, chief executive officer of Ideafarms, an Indian design-in-tech company focused on innovation and social impact.

Vertical farms grow food in such tech-enabled setups and take it to the market through regular supply chains. Over time, they become preferred vendors because their food is organic, of high quality and the supply is predictable. Several tech-enabled vertical farms have proven to be successful. Some examples include Aerofarms and Green Sense in the USA; Delicious in The Netherlands; Sharp's strawberry farm in Dubai; Spread, Toshiba and over 100-plus vertical farms in Japan; and not to forget Packet Greens of Singapore. According to news reports, vertical farming is doing well in Singapore. In the past decade, the number of licensed vertical farms there has risen from one to seven.

Fig. 1: Growing crops vertically in trays at Spread, Japan (Image courtesy: Spread Co. Ltd)

